What is AI?

Artificial Intelligence (AI) is a branch of computer science that focuses on creating machines or software capable of performing tasks that normally require **human intelligence**.

These tasks include:

- Learning from data
- Understanding language
- Recognizing images or speech
- Making decisions
- Solving problems
- Predicting outcomes

Simple definition

AI is the ability of a machine to imitate human intelligence and perform intelligent tasks.

Authentic real-world example

Google Maps uses AI to predict traffic and suggest routes.

For example:

You want to travel from **Indore to Bhopal**.

Google Maps can:

1. Analyze current traffic data.
2. Learn from historical traffic patterns.
3. Predict which roads will be congested.
4. Suggest the fastest route.
5. Recalculate the route if traffic changes.

That's AI because the system is using data to **analyze, predict, and make decisions**.

AI and Its Subsets — Definitions with Examples

1. Artificial Intelligence (AI)

Definition:

Artificial Intelligence is a branch of computer science that aims to create machines or software capable of performing tasks that normally require human intelligence, such as **learning, reasoning, problem-solving, decision-making, perception, and language understanding**.

Example:

A **self-driving car** uses AI to understand its surroundings, recognize objects, follow roads, and make driving decisions.

Real-world comparison:

A human driver observes the road, understands traffic, and decides when to brake. AI enables a computer system to perform similar decision-making tasks.

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2. Machine Learning (ML)

Definition:

Machine Learning is a **subset of AI** in which computers learn patterns from **data** and use those **patterns to make predictions or decisions** without being **explicitly programmed** for every possible situation.

Example:

A spam filter receives thousands of emails labeled:

Email → Spam

Email → Not Spam

Email → Spam

It learns patterns and predicts whether a new email is spam.

Real-world comparison:

A student solves many examples of a particular type of math problem. After seeing enough examples, the student learns the pattern and can solve a new problem.

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3. Deep Learning (DL)

Definition:

Deep Learning is a **subset of Machine Learning** that uses **artificial neural networks** with multiple layers to automatically learn complex patterns from large amounts of data.

Real-world comparison:

A child learns to recognize a cat by seeing many cats. Similarly, a deep-learning model learns visual patterns by processing many images.

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4. Natural Language Processing (NLP)

Definition:

Natural Language Processing is a field of AI that enables computers to **understand, interpret, process, and generate human language**.

Example:

When you type:

"Book a flight from Delhi to Mumbai."

An NLP system can identify:

Source → Delhi

Destination → Mumbai

Intent → Book Flight

Real-world examples:

- Chatbots
- Google Translate
- Voice assistants
- Sentiment analysis
- Text summarization

Real-world comparison:

Just as a human listens to a sentence and understands its meaning, NLP allows computers to process human language.

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5. Computer Vision (CV)

Definition:

Computer Vision is a field of AI that enables computers to **extract information, recognize objects, and understand scenes from images and videos**.

Example:

A phone's face-unlock system captures your face and compares its visual features with the stored representation.

Real-world examples:

- Face recognition
- Object detection
- Medical image analysis
- OCR
- Self-driving cars

Real-world comparison:

Human eyes receive visual information and the brain interprets it. Computer vision performs a similar interpretation using cameras and computational models.

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6. Generative AI

Definition:

Generative AI is a type of AI that can **create new content** such as text, images, audio, video, or code by learning patterns from existing data.

Example:

You give an AI the prompt:

"Write a Python program to sort an array."

It generates Python code based on patterns learned during training.

Other examples:

- Text generation
- Image generation
- Music generation
- Video generation
- Code generation

Real-world comparison:

A human writer learns language and writing patterns from reading many books and can then create a new story. Generative AI similarly learns patterns from data and produces new content.

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7. Neural Networks

Definition:

A Neural Network is a computational model inspired by the human brain that consists of interconnected nodes, called **neurons**, organized into layers to learn patterns from data.

Basic structure:

Input Layer → Hidden Layers → Output Layer

Example:

For handwritten digit recognition:

Image of "7"

↓

Neural Network

↓

Probability

↓

7 = 98%

The model predicts that the image represents the digit **7**.

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8. Reinforcement Learning (RL)

Definition:

Reinforcement Learning is a type of Machine Learning in which an **agent learns by interacting with an environment and receiving rewards or penalties for its actions**.

Basic idea:

Agent

↓

Action

↓

Environment

↓

Reward / Penalty

↓

Learn

Example:

An AI learns to play chess.

- Makes a good move → positive reward
- Makes a poor move → negative reward
- Wins the game → large positive reward

Over many games, it learns strategies that improve its performance.

Real-world comparison:

A child learns to ride a bicycle through trial and error. Successful actions are reinforced, while mistakes help the child adjust.

Agentic AI

Definition

Agentic AI is an AI system that can autonomously pursue a goal by understanding the situation, planning steps, using tools, taking actions, observing the results, and adapting its approach to complete the task.

The key idea is **AI doesn't just give an answer—it can take actions to achieve a goal.**

Simple Flow

Goal
↓
Understand the problem
↓
Plan
↓
Take Action
↓
Observe Result
↓
Adjust / Re-plan
↓
Complete Goal

Example

Suppose you tell an Agentic AI:

"Plan my trip to Mumbai for 3 days."

A normal chatbot might respond with:

Here are some places you can visit...

An **agentic AI system** could potentially:

Your Goal
↓
Search flights
↓
Compare prices
↓
Search hotels
↓
Create itinerary
↓
Check timings
↓
Adjust plan based on constraints
↓
Present final plan

If connected to appropriate tools and given permission, it could also **make bookings or perform other actions**.

Real-World Comparison

Think of the difference between a **consultant** and a **personal assistant**.

Normal AI:

You:

"Find me a good hotel in Mumbai."

AI:

"Here are 5 hotels."

Agentic AI:

You:

"Find me a good hotel in Mumbai under ₹5,000 and arrange my stay."

Agentic system:

Understand budget

↓

Search hotels

↓

Compare options

↓

Select suitable hotel

↓

Check availability

↓

Ask for confirmation if required

↓

Book

So:

Traditional AI → Answers / predicts

Agentic AI → Reasons + plans + acts + adapts

Prompt Engineering — In a Nutshell

Definition

Prompt Engineering is the process of designing and optimizing instructions (prompts) given to an AI model to obtain accurate, relevant, and desired outputs.

Simple Example

Bad prompt:

Tell me about Python.

Better prompt:

Explain Python to a beginner in 5 bullet points, with one real-world example for each point.

The second prompt gives the AI **clear instructions, context, and output format**, so the response is more useful.

Types of Prompting

1. Zero-Shot Prompting

Give the AI a task **without providing examples**.

Example:

Classify this review as positive or negative: "The product is excellent."

Output: Positive

Idea:

No examples → Direct instruction

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2. One-Shot Prompting

Give the AI **one example** before asking it to perform the task.

Example:

"I love this phone." → Positive

"This phone is terrible." → ?

Output: Negative

Idea:

1 example → Learn the pattern

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3. Few-Shot Prompting

Give the AI **multiple examples** to demonstrate the desired pattern.

Example:

"I love this movie." → Positive

"This movie is boring." → Negative

"Amazing acting!" → Positive

"The story was terrible." → ?

Output: Negative

Idea:

Multiple examples → Follow the demonstrated pattern

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4. Role Prompting

Tell the AI to behave as a particular **role or persona**.

Example:

"Act as a Java interviewer and ask me 10 beginner-level interview questions."

The AI adapts its response to that role.

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5. Chain-of-Thought Prompting

Prompting a model to reason through a problem in **multiple steps** rather than immediately giving an answer.

Example:

"Solve this mathematics problem step by step."

This can help with complex reasoning tasks.

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6. Instruction Prompting

Give **clear, specific instructions** about what the AI should do.

Example:

"Summarize this article in exactly 5 bullet points using simple English."

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7. Contextual Prompting

Provide relevant **background information/context** along with the task.

Example:

"You are helping a first-year computer science student. Explain recursion using a real-world analogy and a Java example."

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8. Structured Prompting

Specify the desired **format/structure** of the output.

Example:

"Explain SQL injection using this format: Definition → Example → Prevention."

Quick Revision

Type	**Key Idea**
Zero-Shot	No examples
One-Shot	One example
Few-Shot	Multiple examples
Role Prompting	Assign a role
Chain-of-Thought	Encourage step-by-step reasoning
Instruction Prompting	Clear task instructions
Contextual Prompting	Give background/context
Structured Prompting	Specify output format

Easy Memory Trick

Zero → One → Few = number of examples.

Role → Who

Instruction → What

Context → Background

Structure → Format

Chain-of-Thought → Reasoning